

Interactive Dashboard for Exploring Airbnb Listings in New York City

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Abstract—This project proposes the development of a web-based, interactive data visualization dashboard using D3.js to explore the New York City Airbnb Open Data. As the sharing economy grows, travelers are presented with an overwhelming number of accommodation choices, making manual comparison difficult. The primary goal of this project is to assist budget-conscious travelers in efficiently finding and comparing affordable, high-value accommodation options by balancing location, price, and room type. The proposed visualization will employ coordinated views, including a geospatial map, interactive histograms, and filtering controls. This system aims to reduce the cognitive load of users by allowing them to filter, explore, and analyze listing patterns dynamically.

I. INTRODUCTION

WITH the rapid expansion of the sharing economy, platforms like Airbnb have revolutionized the way people travel, offering diverse accommodation options ranging from private rooms to entire apartments. New York City (NYC), being one of the most popular tourist destinations globally, has a massive number of listings. However, this abundance creates a problem of “choice overload” for travelers.

The specific problem this visualization aims to address is the difficulty budget-conscious travelers face when trying to balance multiple conflicting constraints: location convenience (proximity to the city center), affordability, and accommodation type. Browsing through thousands of listings in a list format is inefficient and fails to reveal spatial pricing patterns (e.g., which neighborhoods offer the best value).

Our target users are tourists and short-term visitors who are unfamiliar with NYC’s geography and rental market. This problem matters because inefficient searching can lead to suboptimal decisions, such as booking an overpriced room or staying in an inconvenient location.

If successful, our visualization tool will provide significant value by enabling users to: 1) Rapidly identify affordable neighborhoods. 2) Filter out irrelevant listings immediately. 3) Make data-driven decisions based on aggregated insights rather than anecdotal evidence.

II. DATA AND DATA PROCESSING

A. Dataset Description

We will use the “New York City Airbnb Open Data” dataset acquired from Kaggle. This dataset provides a comprehensive snapshot of listing activities and metrics in NYC for the year 2019. It contains approximately 49,000 observations and 16 features.

Source: <https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data>

B. Variables of Interest

Table I summarizes the key variables we have selected for visualization. We categorize them into geospatial, numerical, and categorical types to determine appropriate visual encodings.

TABLE I
SELECTED VARIABLES FROM AIRBNB DATASET

Variable Name	Type	Usage in Visualization
latitude	Spatial	Mapping Y-coordinate
longitude	Spatial	Mapping X-coordinate
price	Numerical	Color scale / Filtering
minimum_nights	Numerical	Filtering condition
number_of_reviews	Numerical	Tooltip details / Size
neighbourhood_group	Categorical	Color coding (Regions)
room_type	Categorical	Filtering / Faceting

C. Data Preprocessing

Directly visualizing raw data can be misleading due to noise and outliers. Our preprocessing plan includes:

- 1) **Data Cleaning:** We will remove rows with missing critical values (e.g., null coordinates or prices).
- 2) **Outlier Handling:** Listings with a price of \$0 or extreme outliers (e.g., \$10,000/night) will be filtered out to ensure the color scales in our visualization remain effective.
- 3) **Normalization:** We may derive a new attribute, “Price per Person,” if the data allows, to provide a fairer comparison between different room types.

III. TARGET USER AND TASK REQUIREMENT

Our users need to navigate a complex trade-off between price and location. To support this, we have defined three specific tasks following the visual information seeking mantra: “Overview first, zoom and filter, then details-on-demand.”

- **Task 1 - Spatial Pattern Discovery (Overview):** *Goal:* The user needs to understand the overall price distribution across NYC. *Reason:* Users need to know which boroughs (e.g., Manhattan vs. Brooklyn) are generally more expensive to set realistic budget expectations.
- **Task 2 - Constraint Filtering (Filter):** *Goal:* The user needs to filter listings based on hard constraints: specific price range (e.g., Max \$150) and room privacy (e.g., Private room only). *Reason:* This reduces the search space from 49,000 listings to a manageable subset that fits the user’s wallet.
- **Task 3 - Comparative Analysis (Details):** *Goal:* The user needs to compare specific neighborhoods to find

the "sweet spot"—areas that are relatively cheap but still have high review counts. *Reason:* This helps in identifying hidden gems (high value, low cost) that might be overlooked in a simple list view.

IV. VISUALIZATION DESIGN AND IMPLEMENTATION

We will implement a web-based dashboard using **D3.js** (Data-Driven Documents) to ensure high interactivity and customizability.

A. For Proposal: Design Sketch

Figure 1 illustrates our proposed interface layout. The design is divided into three coordinated views:

- 1) **Main Map View (Left Panel):** This is a scatter plot projected onto a GeoJSON map of NYC.
 - *Visual Encoding:* Each dot represents a listing. The position is determined by latitude/longitude. The color encodes the `neighbourhood_group` or `price` (user selectable).
 - *Interaction:* Pan and zoom capabilities to explore specific streets.
- 2) **Control Panel (Right Top):** A dedicated area for global filters.
 - *Components:* A double-ended slider for the price range and checkboxes for `room_type`.
 - *Feedback:* Changing these inputs immediately filters the points on the map.
- 3) **Statistical View (Right Bottom):** A bar chart or histogram showing the distribution of the currently selected data.
 - *Purpose:* To show the count of listings per neighborhood.
 - *Interaction (Brushing):* Clicking on a bar (e.g., "Queens") will highlight only those listings on the map.

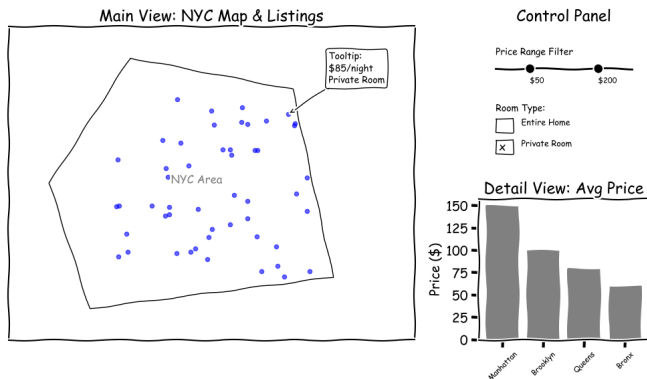


Fig. 1. Low-fidelity sketch of the proposed dashboard. The interface features a map for spatial context (Left) and control panels for filtering attributes (Right).

V. USE CASE

A. For Proposal: Walkthrough

To demonstrate the utility of our design, we present a hypothetical walkthrough of a user named Alex, a student traveling to NYC with a limited budget of \$100 per night.

Step 1: Initial Overview Alex loads the dashboard. The map view shows a dense cluster of 49,000 points. He immediately sees that Manhattan (colored red) is densely packed, while Staten Island has fewer options.

Step 2: Applied Filtering Alex interacts with the **Price Slider**, dragging the maximum handle down to \$100. Simultaneously, he unchecks "Shared Room" because he wants privacy. The dashboard updates in real-time. He observes that most points in lower Manhattan disappear, indicating that area is out of his budget.

Step 3: Exploration Pattern Finding He notices a cluster of remaining points in the *Harlem* neighborhood and *Astoria* (Queens). He glances at the **Bar Chart** on the right, which shows that Astoria has a higher average review count than Harlem for listings under \$100.

Step 4: Details on Demand Intrigued by Astoria, Alex hovers his mouse over a dense cluster of dots in that area. A tooltip pops up displaying: "Private Room, \$75/night, 150 Reviews."

Step 5: Decision Alex confirms that this area offers the best trade-off between price and quality, successfully using the tool to narrow down a massive dataset to a specific, actionable choice.